Evolution in Plant Safety
IEC 61511 Edition 2

Xian Wu, Lifecycle Service (LCS) Business Development Manager

Requirements in order to comply with IEC 61511 ed 2

Generic Information on IEC 61511 ed 2

- IEC 61511 Edition 1 was published in 2003.
- IEC 61511 Edition 2 was published in 2016.
- Comparing with IEC 61511 Edition 1, at IEC 61511 Edition 2, there are more than 200 modifications, highlights which mainly listed as follows:
  - Functional Safety Management System (FSM)
  - Safety Requirements Specification (SRS)
  - Verification and Tests
  - Performance Monitoring
  - Failure Rates / Quantifying Failures
  - New Section on IT Security
  - Bypasses (e.g. MOS)
  - “Grandfather Clause”

© HIMA Paul Hildebrandt GmbH 2017

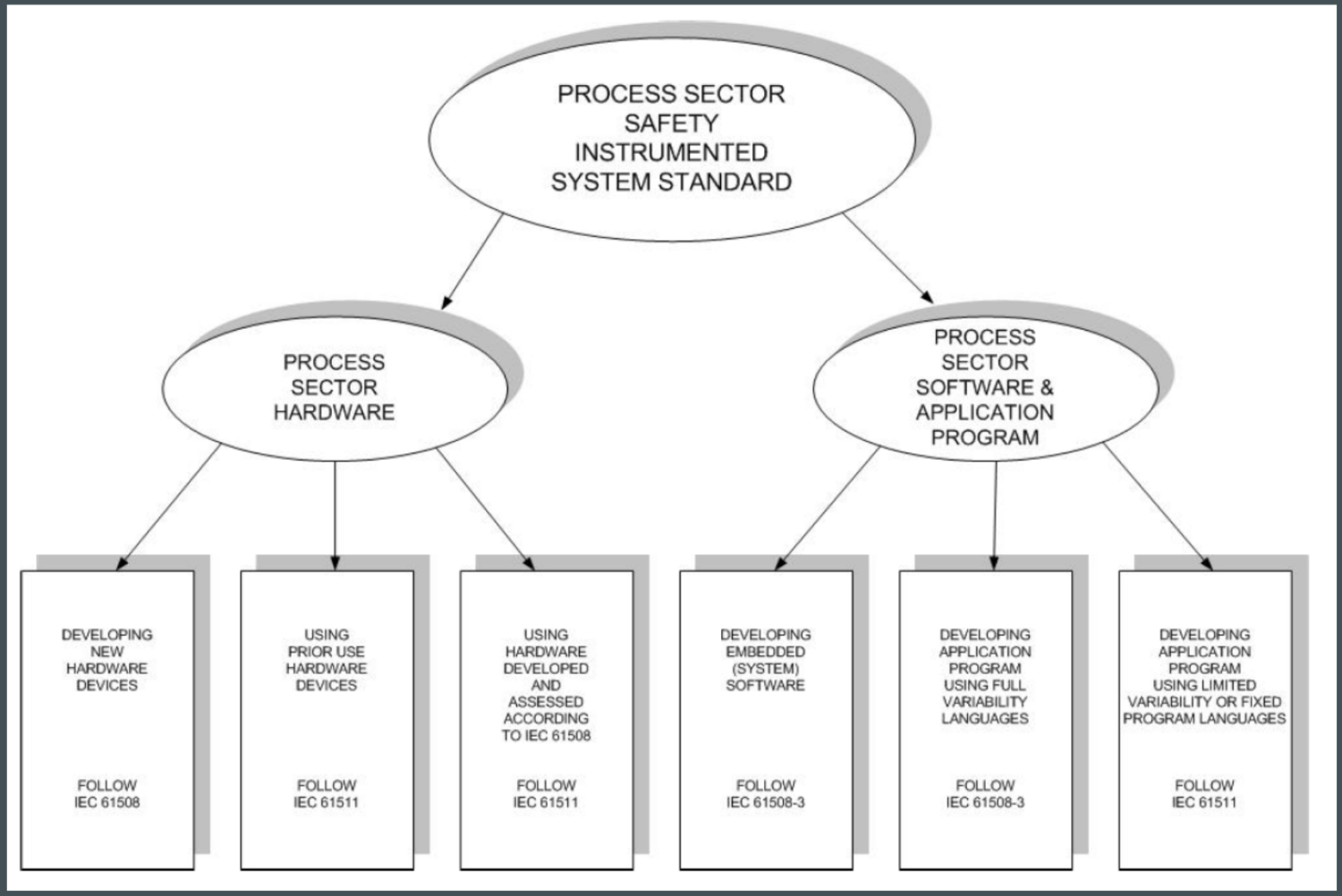
Functional Safety Standards

For Safety Instrumented Systems (SIS) within the process industry sector, there are two important standards when it comes to functional safety:

- IEC 61508 - Functional safety of electrical / electronic / programmable electronic safety-related systems which applies to **Manufacturers and Suppliers of Devices**.
- IEC 61511 - Functional safety: SIS for the process industry sector which applies to **SIS Designers, Integrators and User**.

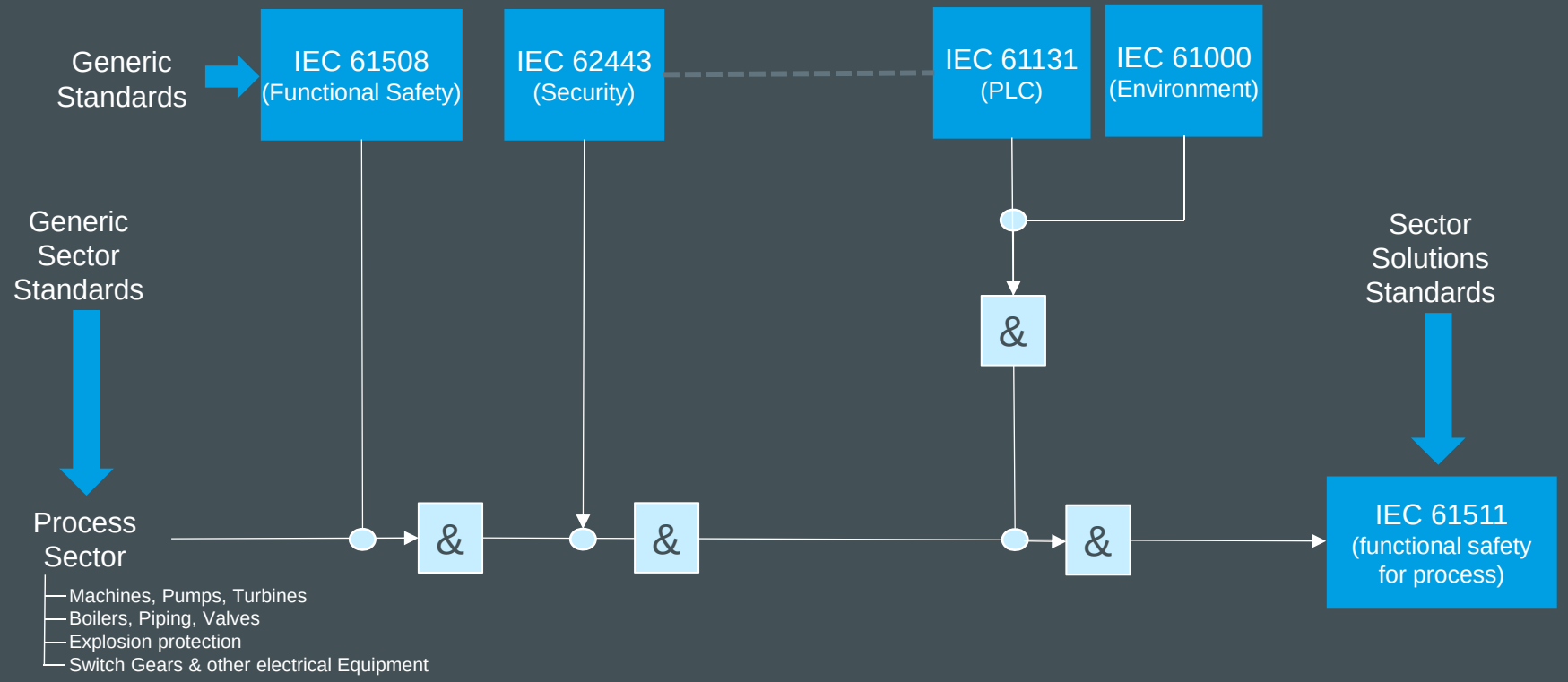
© HIMA Paul Hildebrandt GmbH 2017

Functional Safety: When to look at what?



© HIMA Paul Hildebrandt GmbH 2018

Norms, Standards, correlations & Challenges

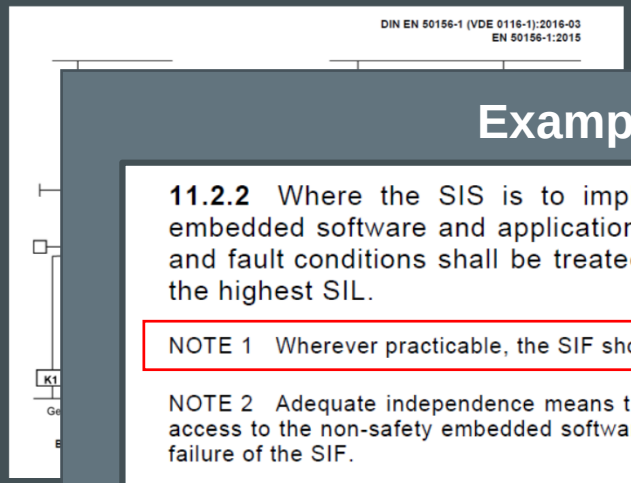


© HIMA Paul Hildebrandt GmbH 2018

Descriptive vs. non descriptive

Descriptive

Non Descriptive



5.2.7SIS configuration management

ng any safety life-cycle
ed:

mented;

s of a device (hardware

service.

utility software (tools)
the application program
under revision control.

Example of interpretations

11.2.2 Where the SIS is to implement both safety and non-SIF then all the hardware, embedded software and application program that can negatively affect any SIF under normal and fault conditions shall be treated as part of the SIS and comply with the requirements for the highest SIL.

NOTE 1 Wherever practicable, the SIF should be physically separated from the non-SIF.

NOTE 2 Adequate independence means that neither the failure of any non-safety functions nor the programming access to the non-safety embedded software or application program functions are capable of causing a dangerous failure of the SIF.

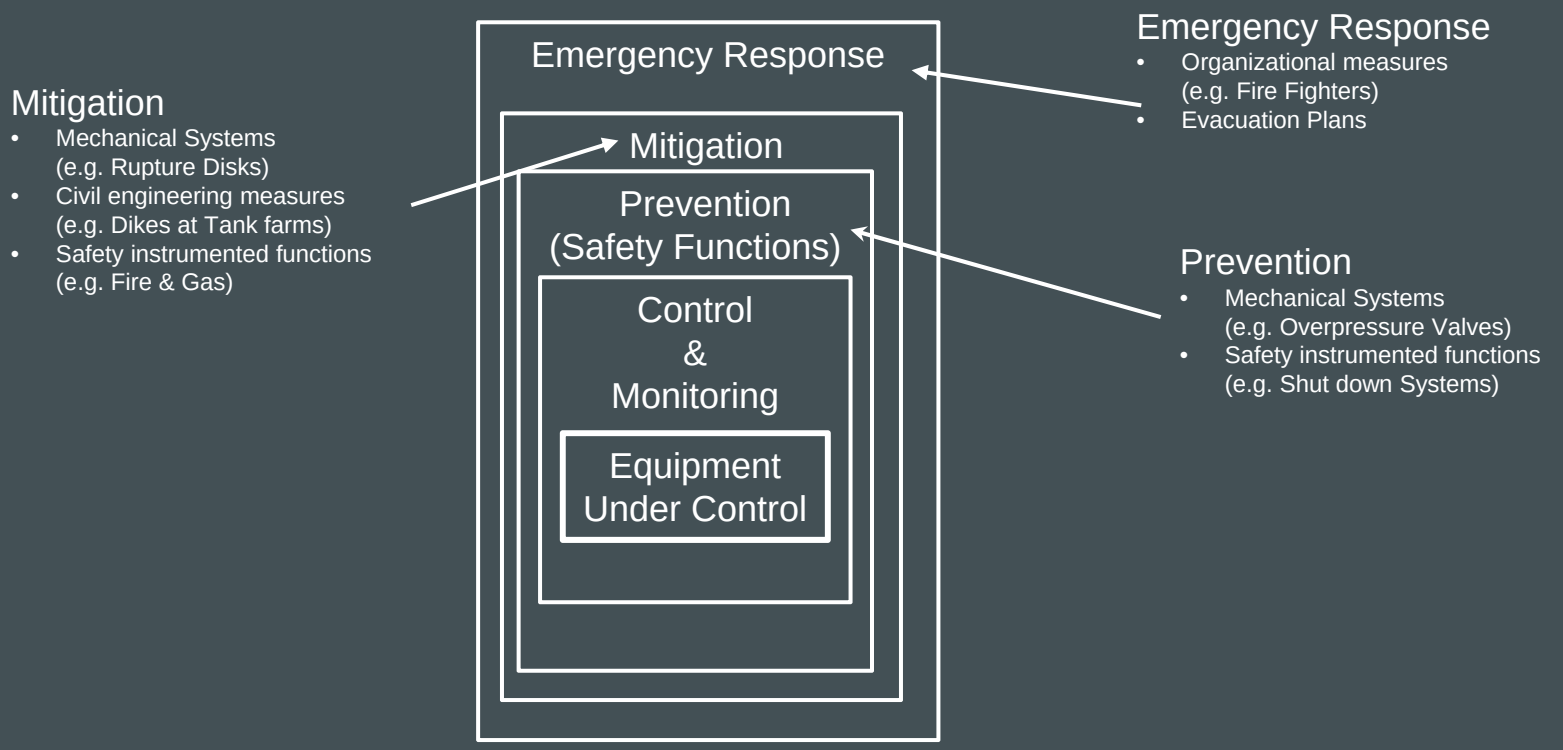
Discussions had recently:

- What is the meaning of ***SHOULD***?
- A Note at a Standard doesn't have normative character
- Shouldn't a logical separation be O.K. too? It is as good as …
- Understood, but an integrated solution can be cheaper, so …

P
C
C
Te
Ir
with others

© HIMA Paul Hildebrandt GmbH 2018

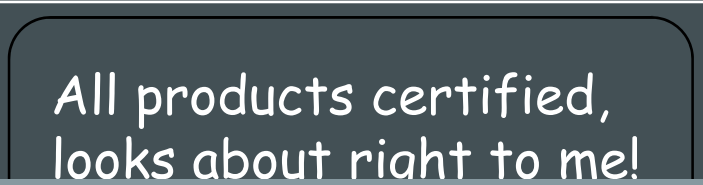
Independent Layer of Protection Concept



What is meant by independence?

© HIMA Paul Hildebrandt GmbH 2018

Compliant plants by certified products?



In order to be compliant you need to

1. Analyze the risk to be mitigated
2. Apply a compliant Design & Engineering process including all recommendations in competency, independence of people involved
3. Test the installation (not just switch it on after installation)
4. Maintain it properly (reliability will drop during operation)
5. Apply an adequate management of change
6. When using certified products, make sure

Thesis: Even if you can prove to use certified products only,
You are not compliant nor safe by definition!

© HIMA Paul Hildebrandt GmbH 2018

The mystery of certification

Example Power Supply:
Input: 230 VAC
Output 24V/20A
SIL rating:
SIL 3 capable Design process
SIL 3 surveillance contact
Isolation Prim/coil: 1,5KV
Isolation Sec/coil: 0,5KV

Would you use this product feeding a safety system?

At least not in an industrial Environment!

Take away

1. Review the safety manuals
2. Check the context of the certification
3. Check the limitations describe
4. Check the environmental conditions

© HIMA Paul Hildebrandt GmbH 2018

Functional Safety Management System

Chapter 5

Requirement:

(5.2) If a supplier makes any functional safety claims for a product or service, the supplier shall have a functional safety management system.

A ‘normal’ quality management system is not enough.

To be covered by FSM:

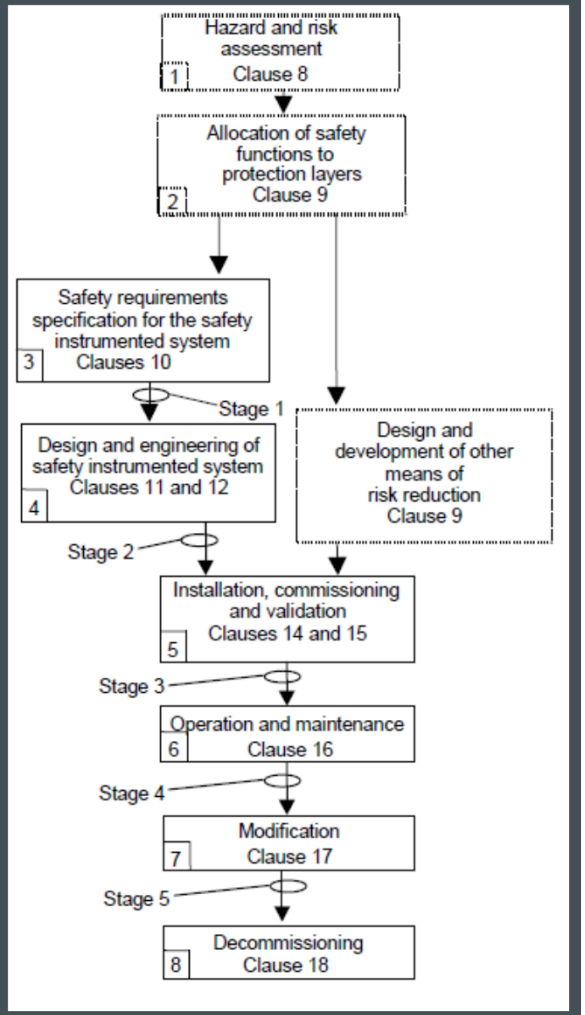
- Organizations & Resources (5.2)
- Risk evaluation & Risk Management (5.3)
- Planning (5.4)
- Implementation & Monitoring (5.5)
- Functional Safety Assessment (5.6)
- SIS Configuration Management (5.7)

© HIMA Paul Hildebrandt GmbH 2017

Lifecycle acc. IEC 61511 Ed. 2

Functional Safety Management System

1. Hazard & Risk Assessment (Chapter 8)
2. Allocation of Safety functions to protection layers (Chapter 9)
3. Safety Requirement Specification (Chapter 10)
4. A compliant engineering & Design process (Chapters 11, 12)
5. A compliant build, installation, commissioning & validation (Chapters 14, 15)
6. A compliant maintenance concept, maintaining the anticipated reliability of the SIS (Chapter 16)



Source: IEC 61511 ed 2

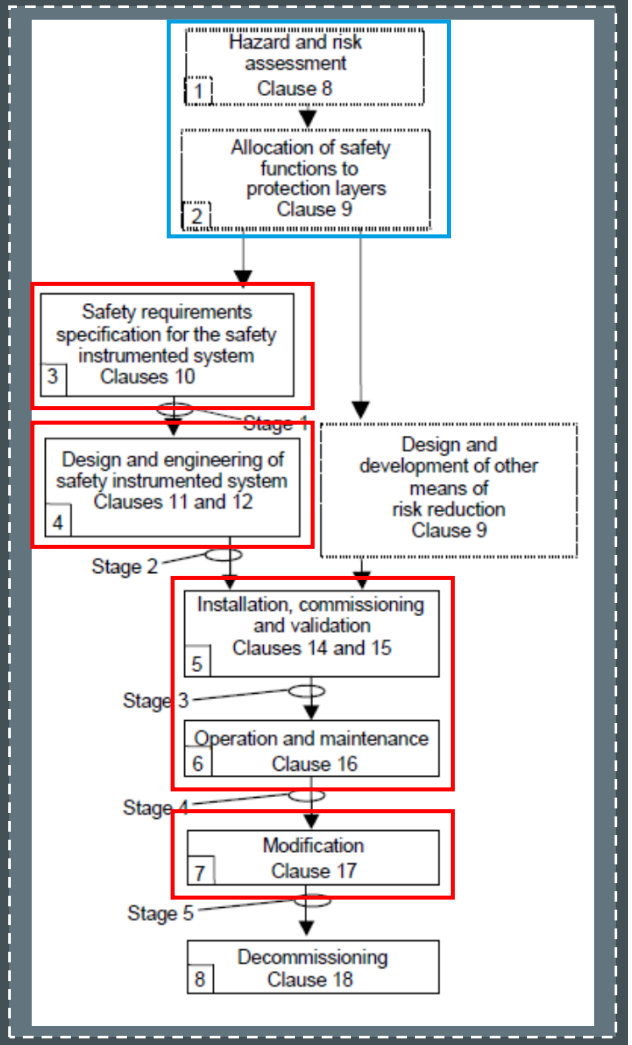
© HIMA Paul Hildebrandt GmbH 2017

HIMA Lifecycle support

Consulting support for Management of functional Safety

Training & Moderation

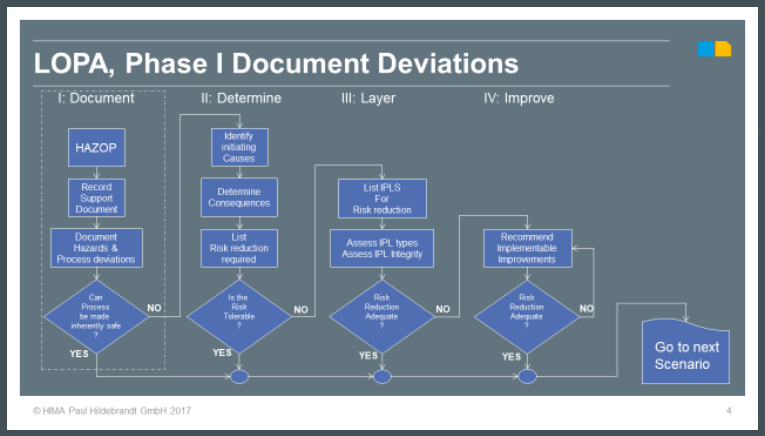
Execution of Tasks

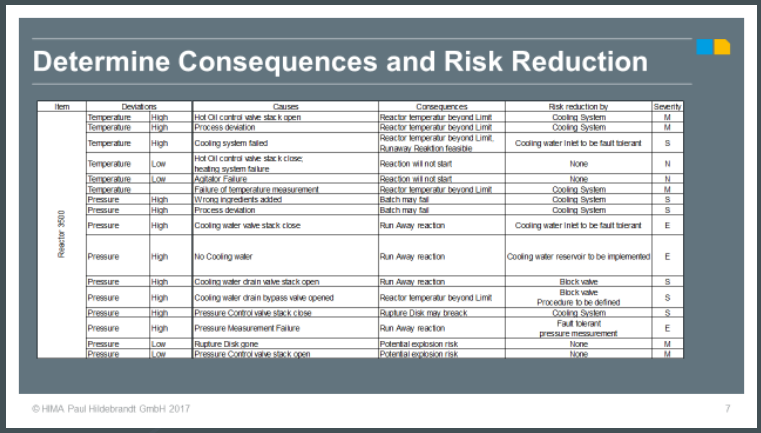


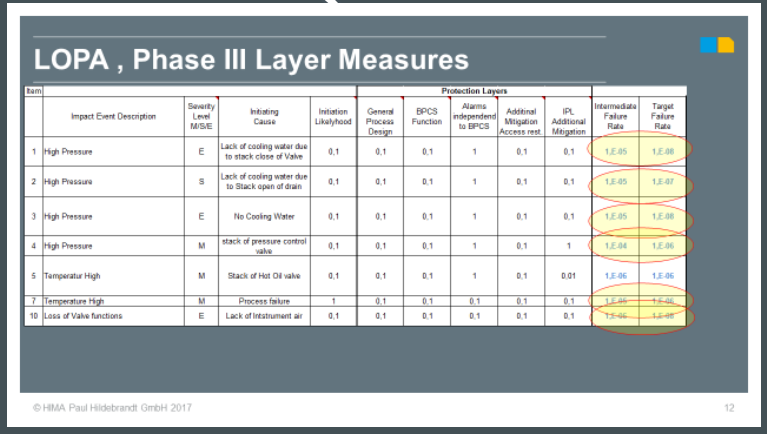
Source: IEC 61511 ed 2

© HIMA Paul Hildebrandt GmbH 2018

Hazard & Risk Assessment & Allocation







Safety Requirement Specification

Safety Requirements Specification (SRS)

Chapter 10

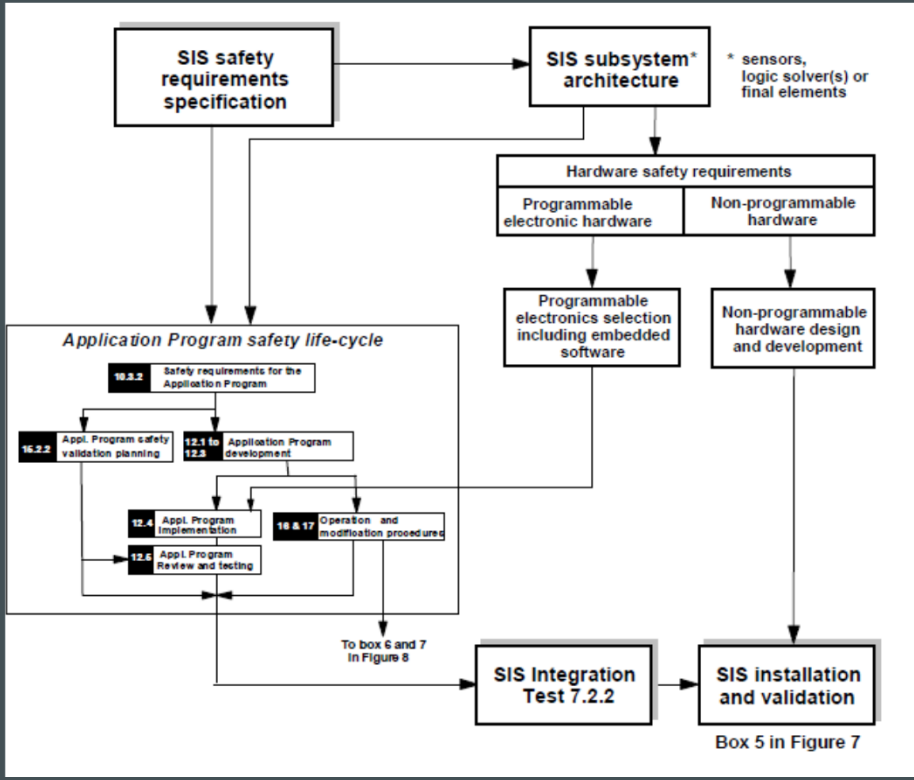
- Clear specification, also in relation to proof test and automated diagnostics
  - SRS shall be understandable, verifiable, testable, modifiable, and traceable
  - Scope, duration, and status of the tested devices, state of the process, detection of common cause failures, and tests from diagnostic devices.
- Requirements for the application program sequencing and time delays (predictability)
- Communication (interfaces, data validation)
- Response time (safety instrumented function response time vs. process safety time)
- Dangerous process states
- Bypassing and related procedures
- Requirements for proof tests and related procedures

© HIMA Paul Hildebrandt GmbH 2017

Safety Lifecycle Activities

Chapter 6

- Structure SIS Architecture
- Define Hardware Components
- Define Hardware Interactions
- Define Software recommendations
- Specify integration tests



© HIMA Paul Hildebrandt GmbH 2018

Some Additional Engineering Highlights

Chapter 11 SIS Design and Engineering

- 6.2.3 Safety planning is a SHALL and includes planning of application programming, with each phase containing: activities, results, criteria, techniques, measures, procedures.

- 11.2.13 A safety manual for operation and maintenance for the specific application shall be written. *The safety manuals of components are only to be consulted as supplementary material or as a reference.*

- SIS design and engineering (chapter 11): When a dangerous fault in a SIS has been detected then compensating measures shall be taken. Where a response of the operator is needed then the alarm shall be considered part of the SIS.

- Normative 13. Factory Acceptance Test (FAT)

© HIMA Paul Hildebrandt GmbH 2017

Verification and Tests

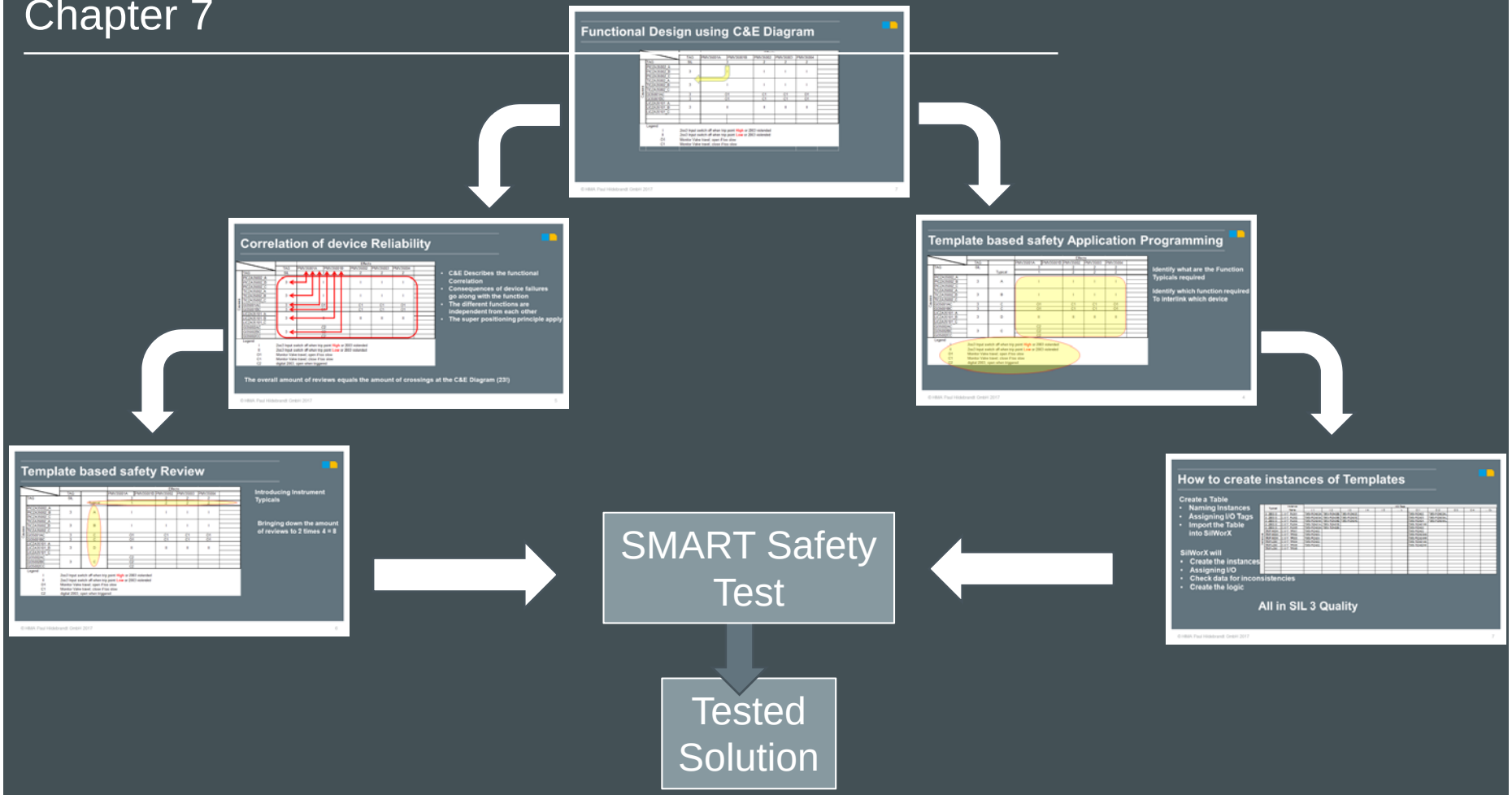
Chapter 7

- Tests are given greater consideration in the context of verification
- New requirements for test planning, e.g.
  - Integration of the application program, hardware, and field devices
  - Integration of subsystems that shall comply with **other standards** (machines or combustion plants)
  - Scope of test (description of test structure, type of test,…)
  - Test environment, including **tools**, hardware, software and their **required configuration**
  - Test criteria (e.g. PASS/FAIL criteria)
  - Procedures for **corrective action on failure** during test
  - Appropriate **personnel**
  - **Management of change**
- Verification for non-interference for non-safety-related functions

© HIMA Paul Hildebrandt GmbH 2017

Verification and Test Support required

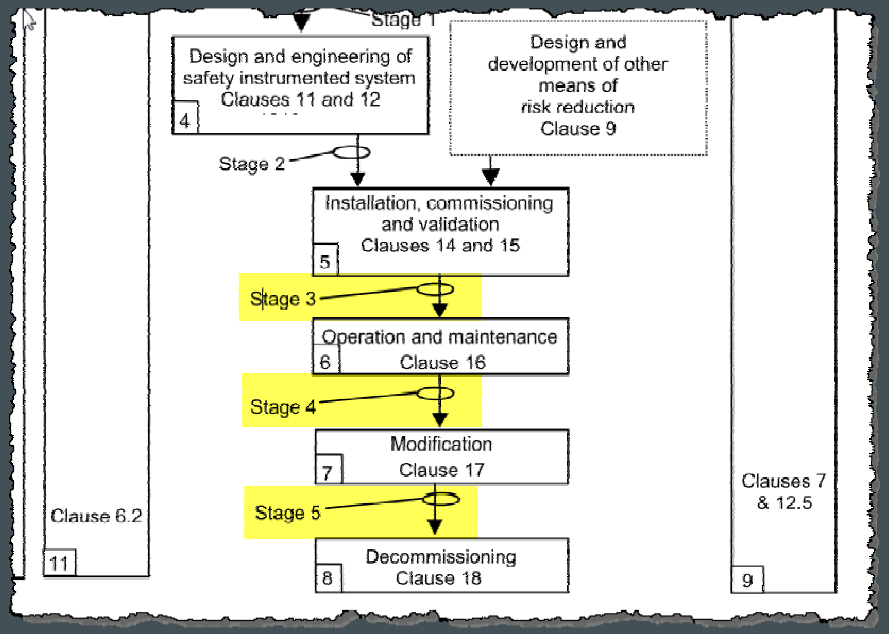
Chapter 7



© HIMA Paul Hildebrandt GmbH 2018

HIMA Lifecycle Support

SIS safety life-cycle and new FSA stages



Mandatory

- Periodical functional safety assessment (FSA) during “Operation & Maintenance Phase” (Stage 4)
- Impact analysis after every modification (Stage 5)

Other Implications

- Newly defined verification and validation (FAT/SAT) test procedures
- Guarantee of competent and trained individuals

© HIMA Paul Hildebrandt GmbH 2017

Performance Monitoring

Chapter 16

- Clarification and additional requirements for monitoring the performance of safety instrumented systems.
  - Monitor and evaluate if the parameters adopted during planning work in actual operational use and determine necessary corrective measures in the case of deviations in:
    - device reliability
    - demand rate for safety instrumented systems
  - Operation and maintenance procedures shall exist for collection and analysis of data regarding the demand rate and reliability parameters of safety instrumented systems.

© HIMA Paul Hildebrandt GmbH 2017

Performance Monitoring

Chapter 16

Consequences of deviating from designed behavior

More frequent demand of the process

than estimated in the risk analysis

→

Low demand rate

changes to

high demand rate

→

Design

e.g. other measures relating to fault tolerance

Operation

e.g. shorter testing intervals

More device failures

than estimated when selecting device

→

Low reliability

leads to higher

likelihood of failures

→

Design

e.g. other devices, automated diagnostics

Operation

e.g. shorter testing intervals

© HIMA Paul Hildebrandt GmbH 2017

Failure Rates / Quantifying Failures

Chapter 11

- The reliability data used when quantifying the effect of random failures shall:
  - be credible, traceable, documented and justified
  - be based on field feedback from similar devices used in a similar operating environment
  - be calculated with an upper bound confidence of 70%

61511 Prior Use

- End-user responsibility
- Field experience, based on comparable applications (documented history)
- Precise requirements and preconditions
- Management of change required
- e.g. NE 130 (min. 100,000 h)

61508 Proven in Use

- Manufacturer responsibility
- Returns must be evaluated (documented history)
- min. 300,000,000 h
- Management of change is required

© HIMA Paul Hildebrandt GmbH 2017

New Section on IT Security

Chapter 8

- Users must take system and network vulnerabilities into account.
- A risk assessment regarding security and access protection has to be performed to identify security vulnerabilities of the Safety Instrumented System (SIS).
  - List of devices covered by this risk assessment (e.g. SIS, BPCS or any other device connected to the SIS)
  - Description of identified threats that could exploit vulnerabilities and result in security events.
  - Consideration of all life-cycle phases in the cybersecurity assessment.
- References to additional security standards:
  - ISA TR84.00.09, ISO/IEC 27001:2013, and IEC 62443-2-1:2010
- The responsibility for defining the conditions for performing an IT risk analysis is typically with the operator of the facility and not with the supplier or manufacturer.

© HIMA Paul Hildebrandt GmbH 2017

Bypasses (e.g. MOS)

Chapter 16

- New requirements relating to bypassing Safety Instrumented Systems and the necessary compensating measures.
- All bypasses (MOS) require:
  - authorization
  - indication
  - status record (bypass log)
  - maximum time allowed
  - compensating measures that ensure continued safety
  - operating procedures for the operator
    - procedures to be applied before and during bypass and what should be done before the removal of the bypass. Definition of the maximum allowed time for a bypass.

© HIMA Paul Hildebrandt GmbH 2017

“Grandfather Clause”

Chapter 5

- A kind of “grandfather clause” is included in the standard.
- There are no compulsory changes to existing safety instrumented systems, but …
  - for existing safety instrumented systems that correspond to regulations, standards, or procedures that were valid prior to the release of the new standard, the user must prove that the existing system is safely:
    - designed
    - maintained
    - inspected
    - and operated.

© HIMA Paul Hildebrandt GmbH 2017

Closing Words

“Love all, trust a few, do wrong to none.”

William Shakespeare

Functional Safety is a complex, important undertaking.

There are no cost of safety reported, but only cost of unsafe behavior getting widely discussed.

© HIMA Paul Hildebrandt GmbH 2018

Thank You!

HIMA Paul Hildebrandt GmbH

Albert-Bassermann-Str. 28
68782 Brühl, Germany

Phone: +49 (0) 6202 / 709-0
Fax: +49 (0) 6202 / 709-107

E-Mail: info@hima.com
Internet: www.hima.de

© HIMA Paul Hildebrandt GmbH 2017